HYDRAULIC VARIABLE VALVE TIMING ACTUATOR [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (WITH 4-1 EXHAUST SYSTEM)]

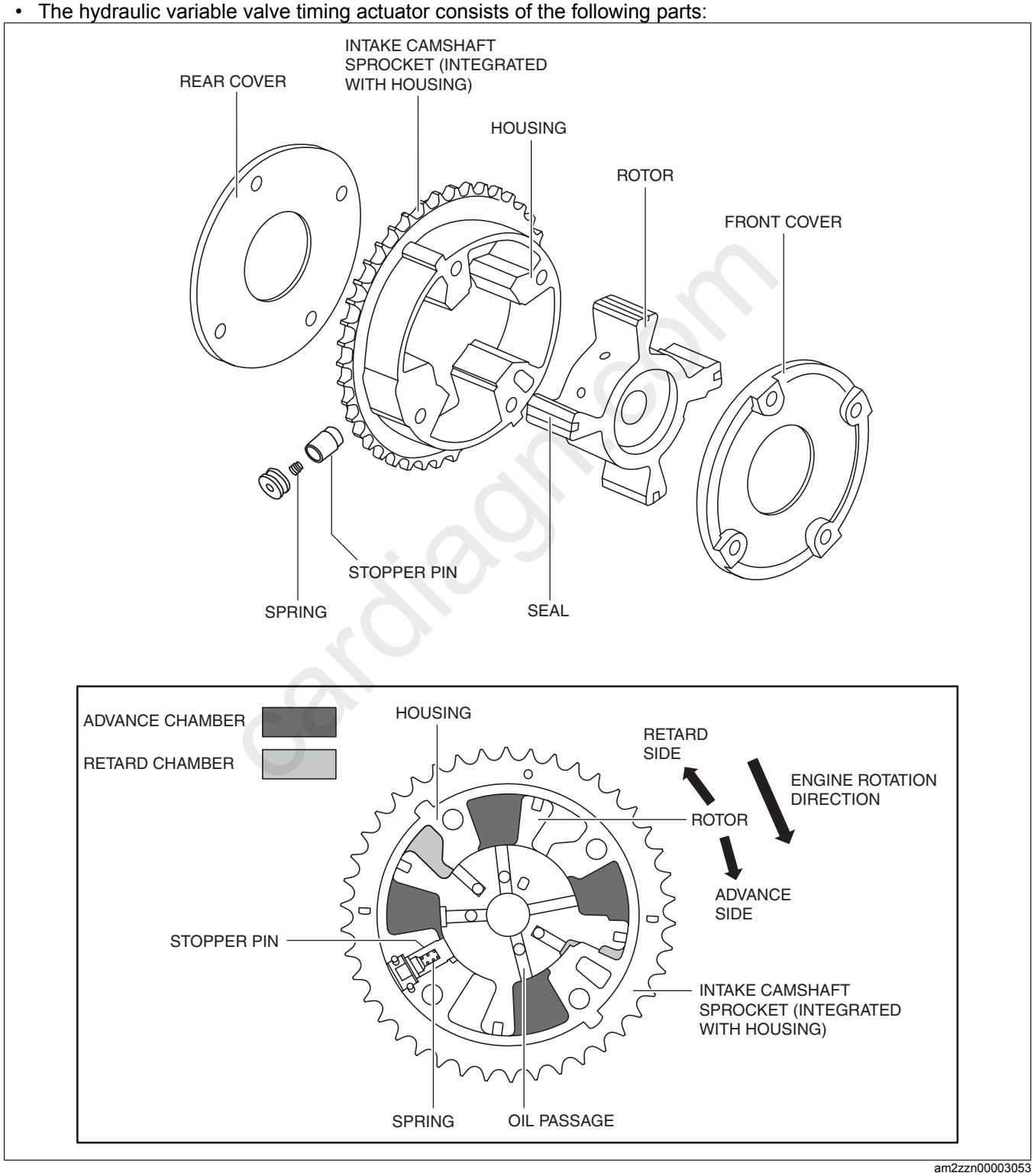
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Purpose, Function

- The hydraulic variable valve timing actuator operates using the engine oil supplied by the OCV, and changes the phases of the intake camshaft (valve timing).
- The hydraulic variable valve timing actuator cannot be disassembled because it is a precision unit.

Construction

- The hydraulic variable valve timing actuator consists of the following parts:



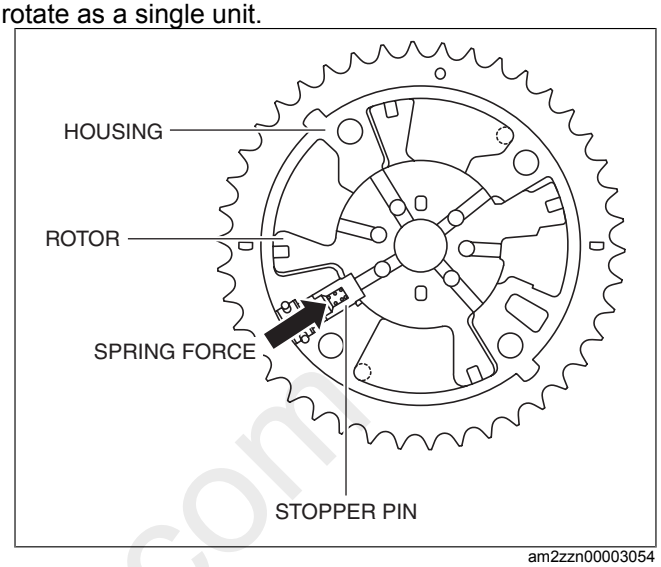
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— The intake camshaft sprocket is integrated with the housing.
— The rotor and the intake camshaft are engaged and they rotate together.
— There are two types of hydraulic chambers; advance chamber and retard chamber, between the housing and the rotor.

— When the rotor (intake camshaft) is in the maximum retard position, the stopper pin is engaged with the rotor depression.

Operation
Fixed mode

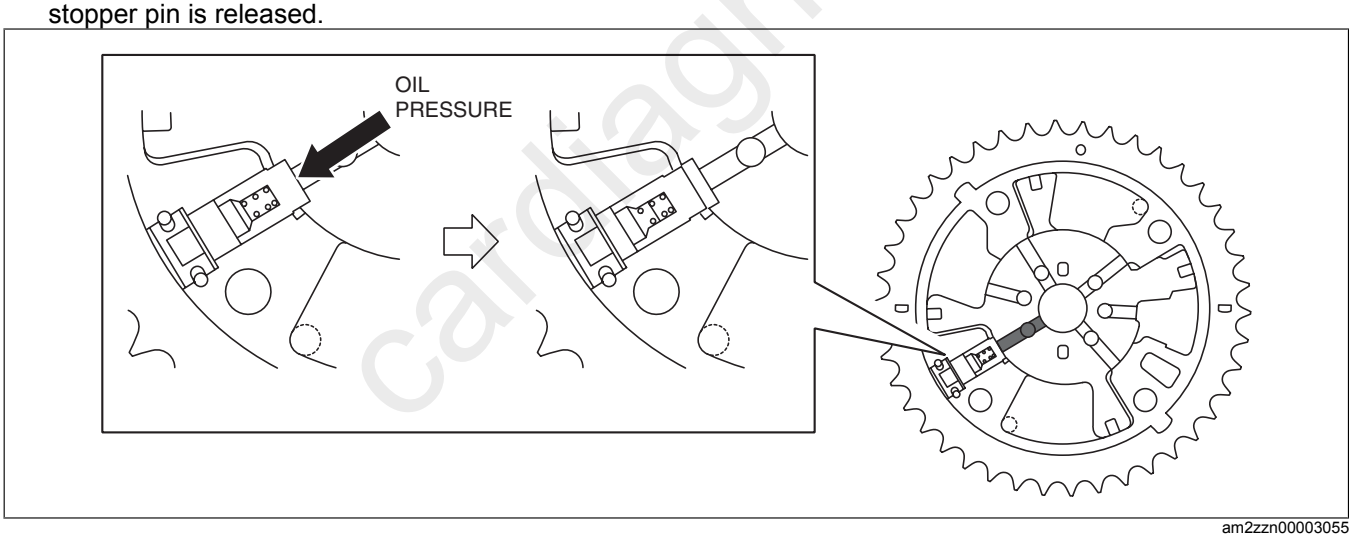
- The stopper pin inside the variable valve timing actuator is engaged with the depression of the rotor positioned in the maximum retard position by the spring force.
- The intake camshaft sprocket and the intake camshaft rotate as a single unit.



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Advance mode

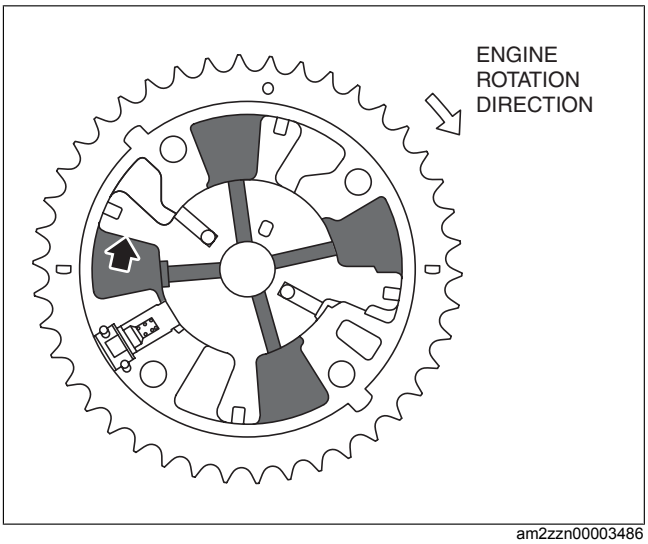
- Engine oil is led to the oil passage by the OCV, and the oil pressure is applied to the stopper pin. As a result, stopper pin is released.



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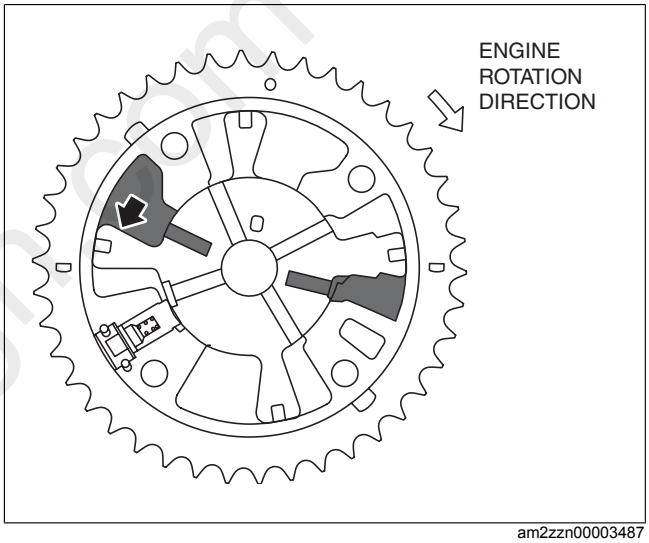
- When the stopper pin is released, oil pressure is applied to the advance chamber. As a result, the rotor rotates in the advance direction relative to the intake camshaft sprocket, and the valve timing is advanced.



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Retard mode

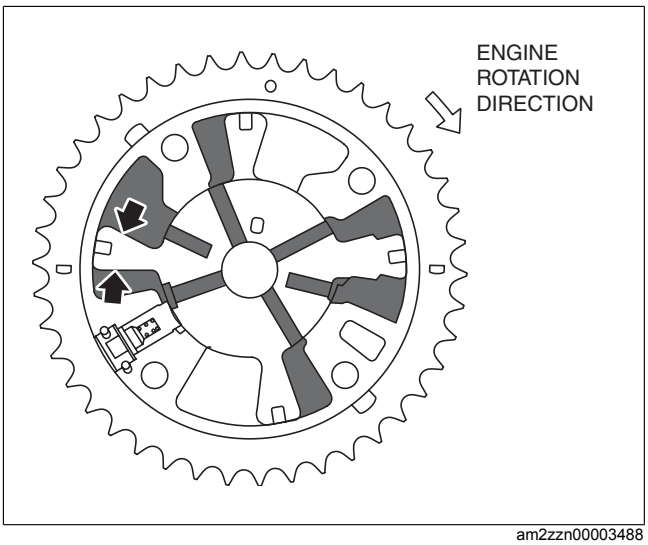
- The oil passage is switched by the OCV and the oil pressure is applied to the retard chamber. As a result, the rotor rotates in the retard direction relative to the intake camshaft sprocket, and the valve timing is retarded.



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Hold mode

- The oil pressure is applied from the OCV to both the advance and retard chambers. As a result, the relative angle of the rotor and the intake camshaft sprocket is not changed and the valve timing is held.



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